



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, BASAR
Basar, Nirmal (Dist), Telangana - 504107

(A.Y. 2025-2026)

Branch: Civil Engineering

R22

Subject Name: Strength Of Materials

Date: 08/08/25

Subject Code: CE2103

Exam: E2S1MT1

Student ID

Time: 45 Minutes

No:

Max. Marks: 15M

Note:

- Answer all questions from Section -A. Each question carries 1 mark.
- Answer any two questions from Section - B. Each question carries 5 marks.
- If data is found to be missing, assume suitable necessary data.

Section-A

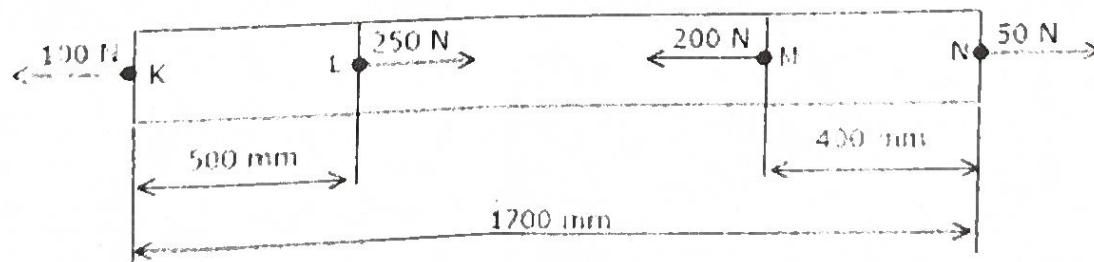
5×1=5M

- Give the note on 'MALLEABILITY' _____? 1M
- If the value of modulus of rigidity is half of the modulus of elasticity then the value of Poisson's ratio is _____? 1M
- The number of independent constants for a Orthotropic material is _____? 1M
- A Hopper section having diameter top is 30mm and bottom diameter is 20mm is subjected to a tensile load of 45kN for a length of 3m section. Calculate the deformation? 1M
- Draw the stress - strain graph for a brittle material _____? 1M

Section-B

2 × 5 = 10M

- A short column is having a diameter is of 80mm and it is subjected to compressive load 80kN, then the value of change in diameter is _____? 3M
Note: $\mu=0.25, E=1 \times 10^5 \text{ N/mm}^2$
- A steel rod 10 mm in diameter and 1 m long id heated from 20°C to 120°C , $E=200\text{GPa}$ and $\alpha= 12 \times 10^{-6}\text{per}^\circ\text{C}$. If the rod is not free to expand, the thermal stress developed is _____? 2M
- The figure below shows a steel rod of 25mm^2 cross sectional area.it is loaded at four points K,L,M,N. Assume $E= 200\text{GPa}$. The total change in length of the rod due to loading is _____? 3M



- Draw stress, strain graph for mild steel member with correct notations? 2M
- A beam element is having mutual perpendicular stresses and subjected to shear stress. Derive the expression for normal stress,tangential stress if an oblique plane makes an angle of inclination θ ? 5M



Subject Name: Strength Of Materials

Subject Code: CE2103

Date: 25/09/25

Time: 45 Minutes

Exam: E2S1MT2 Student ID No: B220367

Max. Marks: 15M

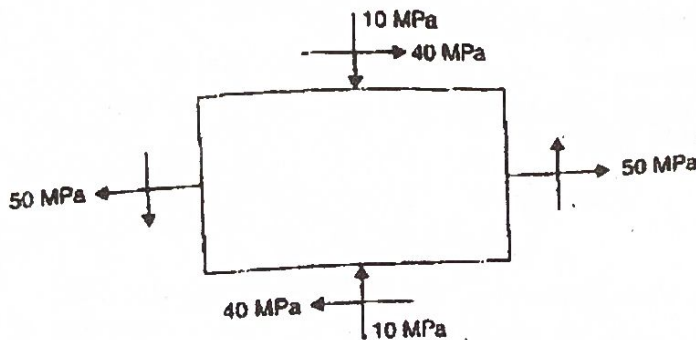
Note:

- Answer all questions from Section -A. Each question carries 1 mark.
- Answer any two questions from Section - B. Each question carries 5 marks.
- If data is found to be missing, assume suitable necessary data.

Section-A

5×1=5M

- The volumetric strain in case of a thin cylindrical shell of diameter d , thickness t , subjected to internal pressure p is.....? 1M
- For the below given element calculate the maximum and minor principal stresses are? 1M



- The diameter of shaft A is twice the diameter of shaft B and both are made of the same material. Assuming both the shafts to rotate at the same speed, the maximum power transmitted by B is.....? 1M
- The ratio of circumferential stress to longitudinal stress in a thin cylinder subjected to internal hydro static pressure is.....? 1M
- A shaft is having diameter 500mm and is subjected to a bending moment $M = 400$ N.m and torque $T = 300$ N.m. Then the combined stress along Principal major stress is.....? 1M

2 × 5 = 10M

Section-B

- In a torsion test, the specimen is a hollow shaft with 50 mm external and 30 mm internal diameter. An applied torque of 1.6 kN-m is found to produce an angular twist of 0.4° measured on a length of 0.2 m of the shaft. The Young's modulus of elasticity obtained from a tensile test has been found to be 200 GPa. Find the values of
(i) Modulus of rigidity.
(ii) Poisson's ratio. 5M
- Determine the maximum and minimum hoop stress across the section of a pipe of 600 mm internal diameter and 200 mm thick, when the pipe contains a fluid at a pressure of 9.5 N/mm^2 . Also sketch the radial pressure distribution and hoop stress distribution across the section? 5M
- a) A cylindrical shell has the following dimensions: 4M
Length = 3 m, Inside diameter = 1 m, Thickness of metal = 10 mm
Internal pressure = 1.5 MPa
Calculate the change in dimensions of the shell and the maximum intensity of shear stress induced. Take $E = 200 \text{ GPa}$ and Poisson's ratio $\nu = 0.3$
b) Two steel shafts, one solid of diameter D and the other hollow of outside diameter D and inside diameter $D/2$, are twisted to the same angle of twist per unit length. The ratio of maximum shear stress in solid shaft to that in the hollow shaft is..? 1M



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Branch: Civil Engineering

R22

Subject Name: Strength Of Materials

Date: 31/10/25

Exam: E2S1MT3

Student ID No: B220367

Subject Code: CE2103

Time: 45 Minutes

Max. Marks: 15M

Note:

- Answer all questions from Section -A. Each question carries 1 mark.
- Answer any two questions from Section - B. Each question carries 5 marks.
- If data is found to be missing, assume suitable necessary data.

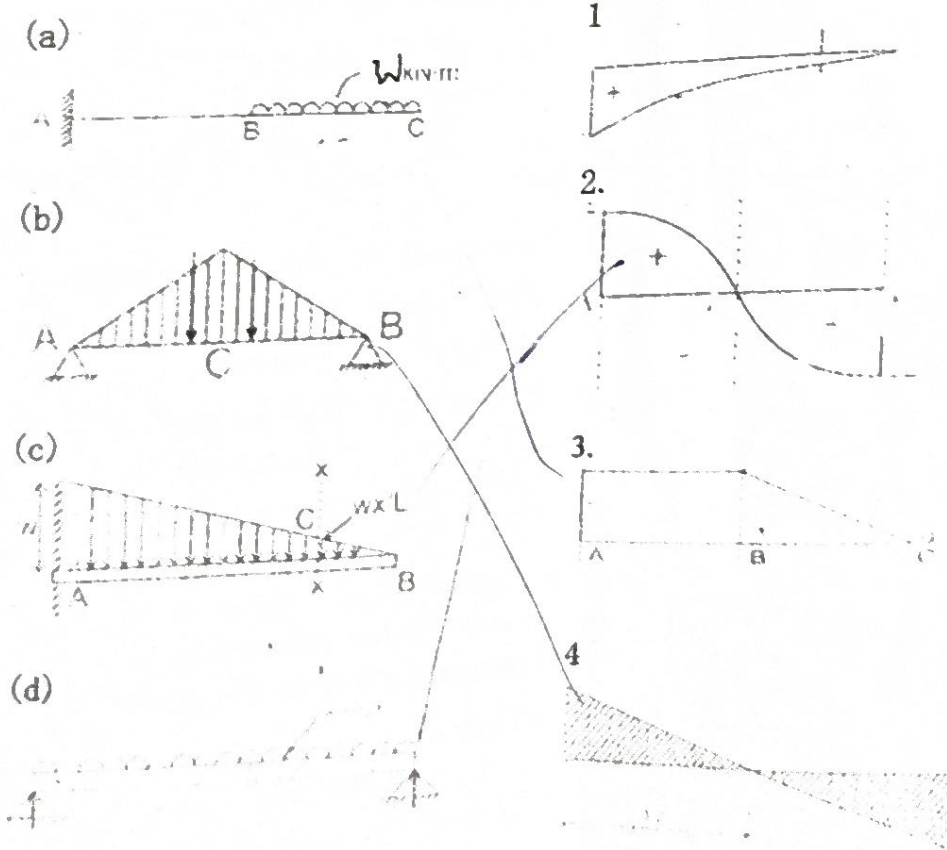
Section-A

5×1=5M

- The value of maximum bending moment for a simply supported beam carrying a load whose intensity varies uniformly from zero at one end to w per unit run at the other end is _____? 1M
- Draw the shear stress distribution diagram for a 'T' section.? 1M
- Write the Rankine's constant 'a' value from the crippling load relation? 1M
- A steel wire is having a diameter of 20 mm, bent into a circular shape with 10 m radius. If $E = 2 \times 10^5 \text{ N/mm}^2$, then $\sigma_{\max} = ?$ 1M
- Match the following: 1M

Beam

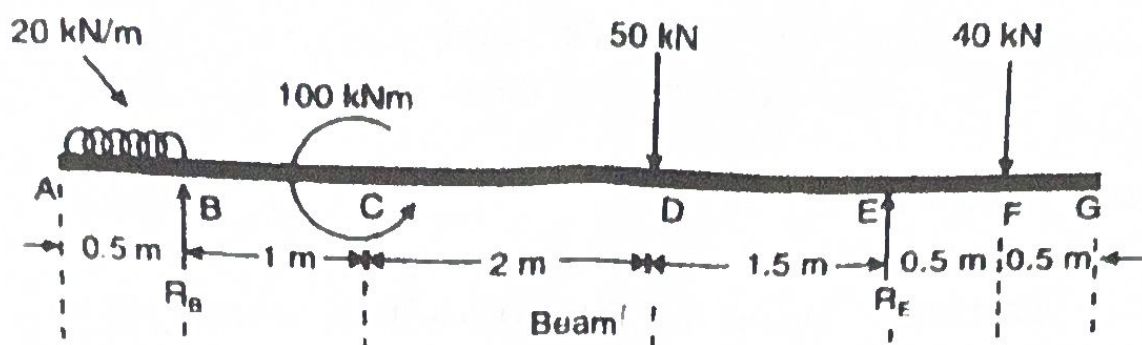
Shear-force Diagram



Section-B

2 × 5 = 10M

6. Construct the bending moment and shearing force diagrams for the beam shown in the figure. 5M



7. A simply supported beam of span 1.3 m having a cross section 150 mm wide by 250 mm deep carries a concentrated load W at the centre. Allowable working stresses are: $\sigma = 7 \text{ N/mm}^2$ (bending), shear stress $\tau = 1 \text{ N/mm}^2$ (shear). What is the safe load W ? 5M
8. Calculate the safe compressive load on a hollow cast iron column (one end rigidly fixed and the other hinged) of 150 mm external diameter and 100 mm internal diameter and 10 m in length. 5M
Use Euler's formula with a factor of safety of 5.
 $E = 95 \text{ kN/mm}^2$.

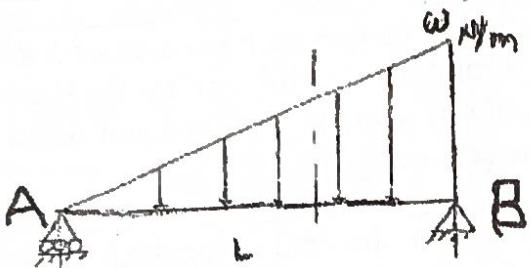
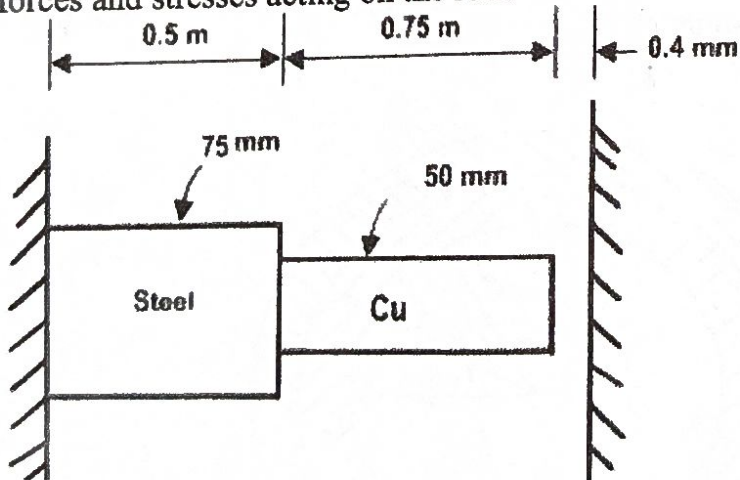
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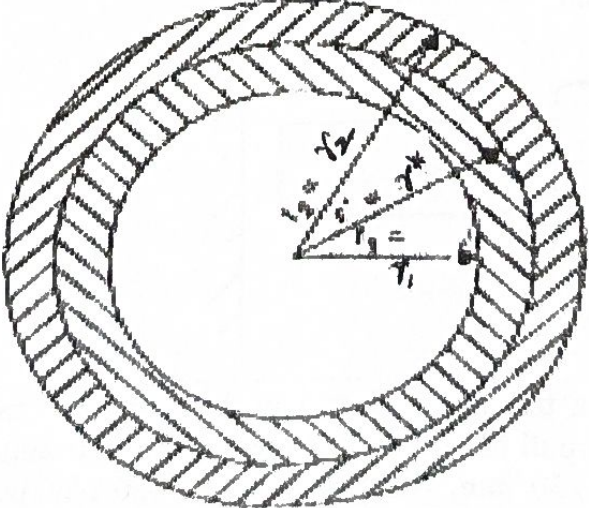
	RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, BASAR Basar, Nirmal (Dist), Telangana - 504107	
	(A.Y. 2025-26) Branch: Civil Engineering	
		R22

Subject Name:	Mechanics of Materials	Subject Code:	CE2103
Date:	19/11/25	Time:	180 Minutes
Exam:	E2S1EST_REM	ID No:	
		Max. Marks:	60

5×12=60Marks

Answer any five of the following questions.

All the answers to the sub questions must be written together. If any data found to missing, assume necessary suitable data.		
1. a)	For the below given beam member calculate the values of shear, bending moment and also draw the SF, BM diagrams to the corresponding loading? 	6M
b)	A rod consists of two parts that are made of steel and copper as shown in figure below. The elastic modulus and coefficient of thermal expansion for steel are 200 GPa and 11.7×10^{-6} per $^{\circ}\text{C}$ respectively and for copper 70 GPa and 21.6×10^{-6} per $^{\circ}\text{C}$ respectively. If the temperature of the rod is raised by 50°C, determine the forces and stresses acting on the rod? 	6M
2. a)	In a tensile test, a test piece 30 mm in diameter, 250 mm gauge length stretched 0.106mm under a pull of 60 kN. In a torsion test, the same rod twisted 0.017 radian over a length of 250 mm, when a torque of 450 Nm was applied. Find the three elastic moduli and the Poisson's ratio for the material of the test piece?	10M
b)	Euler's critical load for a column with both ends hinged is found as 40 kN. What would be the change in the critical load if both ends are fixed?	2M

3. a)	Two planes AB and BC which are at right angles are upon by tensile stress of 140 N/mm^2 and a compressive stress of 70 N/mm^2 respectively and also by shear stress 35 N/mm^2 . determine the principal stresses and principal planes. Find also the maximum shear stress and planes on which they act?	10M
b)	The volumetric change in volume of a thin cylinder under internal pressure having hoop stress = 200 MPa , $E = 200 \text{ GPa}$ and Poisson's ratio = 0.25 is?	2M
4. a)	A beam AB, 20 meters long supported on two intermediate supports 12 meters apart carries a uniformly distributed load 6 kN/m together with concentrated loads of 30 kN at the left end A and 50 kN at the right end B. The supports are so located that reaction in the same of each support. Determine the position of the supports and draw B.M and S.F diagrams. Mark the values of the B.M and S.F ?	10M
b)	A hollow shaft of outer dia 40 mm and inner dia of 20 mm is to be replaced by a solid shaft to transmit the same torque at the same maximum stress. What should be the diameter of the solid shaft?	2M
5. a)	A short column of rectangular cross - section 80 mm by 60 mm carries a load of 40 kN at a point 20 mm from the longer side and 35 mm from the shorter side. Determine the maximum compressive and tensile stresses in the section?	10M
b)	Write the general equations of slope and deflection for a beam with uniform load conditions?	2M
6. a)	A simply supported beam AB of span length 6 m carrying a point load of 3 kN at a distance of 2 m from each end. If $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 10^8 \text{ mm}^4$ for the Beam AB, then using conjugate beam method determine: i) slope at each end and under each load ii) deflection under each load and at the centre?	10M
b)	Draw the shear stress distribution diagrams for a circular, rectangular, I- section, T- section and write any two shear stress values up to neutral axis depth ?	2M
7.	A compound cylinder is made by shrinkage a cylinder of external diameter 350 mm and internal diameter of 300 mm over another cylinder of external diameter 300 mm and internal diameter 250 mm. The radial pressure at the junction after shrinkage is 8 N/mm^2. Find the final stresses set up across the section, when the compound cylinder is subjected to an internal fluid pressure of 84.5 N/mm^2 ? 	12M